SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning
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COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Scenario-Based Assessment **DURATION: 1.5 Hrs**

A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and suitable learning algorithm. Ans: -

Introduction
A smart home can use Reinforcement Learning (RL) to automatically control electrical appliances such as lights, fans, and air conditioners. The RL agent observes the user's presence, room temperature, weather conditions, and electricity consumption. Based on this information, it learns when to switch appliances ON or OFF to reduce energy consumption while maintaining user comfort.

1. State Space
The state represents the current condition of the smart home.
The state can include:

- User presence: Present / Absent
- Room temperature: Low / Normal / High
- Weather condition: Hot / Normal / Cold
- Current electricity consumption: Low / Medium / High Time of the day: Morning / Afternoon / Night

Example:
State = (Presence, Temperature, Weather, Power Consumption)
For example, if the user is present, the temperature is high, and power consumption is low, the agent may decide to turn ON the fan or AC.

2. Action Space
The actions represent decisions made by the RL agent.
Possible actions are:
- Turn light ON/OFF
- Turn fan ON/OFF
- Turn AC ON/OFF
- Increase or decrease AC temperature
- Keep appliances unchanged

The agent selects an action based on the current state.

3. Reward Structure
The reward should encourage energy saving and user comfort.
A simple reward function can be:
Reward = Comfort Level - Energy Consumption

Examples:
- Low energy consumption + comfortable user → Positive reward
- High electricity consumption → Negative reward
- User uncomfortable → Negative reward
- Appliance ON when nobody is present → Large negative reward Thus, the agent learns to avoid unnecessary electricity usage.

4. Suitable Learning Algorithm
Q-Learning can be used when the states and actions are small and discrete.
For a more complex smart home with continuous values such as exact temperature and power consumption, Deep Q-Network (DQN) or another deep RL algorithm can be used.

5. Working of the RL System
Smart Home Environment
↓
Observe State
↓
RL Agent
↓
Select Action
↓
Control Lights/Fan/AC

↓
Observe New State
↓
Calculate Reward
↓
Update Policy
↓
Repeat Learning

Example Suppose:
User = Present
Temperature = High
Weather = Hot Power
= Medium The agent
may choose:
AC → ON
Fan → ON Light
→ OFF

If the user becomes comfortable without excessive energy consumption, the agent receives a positive reward.
If the user leaves the room, the agent can learn to turn OFF the AC and lights, reducing unnecessary energy consumption.

Advantages

1. Reduces electricity consumption.
2. Maintains user comfort.
3. Automatically adapts to user behavior.
4. Avoids unnecessary appliance usage.
5. Can learn optimal energy-management policies over time.

Conclusion

The proposed RL-based smart home system can intelligently control appliances according to user presence, temperature, weather, and energy consumption. By providing rewards for energy savings and comfort, the agent learns an efficient control policy. Therefore, Reinforcement Learning is suitable for developing an automated and energy-efficient smart home system

Code of Conduct:

I <u>Mahendra Reddy (192425214)</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand